

Pierre Fihey

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Education

- Institut Polytechnique de Paris**, PhD in Computer Science 2025 – ...
- Exploring the geometry of textual representations and the use of statistical tools such as data depth functions for Anomaly Detection.
- Institut Polytechnique de Paris**, Master 2 Data Sciences 2023 – 2024
- Optimal Transport, Advanced AI for texts and graphs, NLP, Audio and music information retrieval, Optimization for data science...
- Paris Dauphine University**, Master 2 Statistical and Financial Engineering (Apprenticeship program) 2022 – 2023
- *Master thesis*: Financial news studies to automatize investments using NLP Algorithms.
- Paris Dauphine University**, Master 1 Applied Mathematics 2021 – 2022
- *Master thesis*: Prediction of corporate fraud using multiple Machine Learning tools.
- Paris Dauphine University**, Bachelor's degree in Applied Mathematics 2017 – 2021

Experience

- Adjunct Lecturer**, Univesrity Paris Dauphine - Master 2 Statistical and Financial Engineering 2024-...
- *Course* : Python for Data Science.
 - Development of course content in unsupervised and supervised learning, deep learning, and NLP, including a project on automated decision-making in financial markets.
- Research Intern in NLP**, Telecom Paris - LTCI 2024-2025
- Study of Large Language Models for Anomaly Detection in text data: Unsupervised Learning, Knowledge Distillation, Contrastive Learning, Representation Learning..
- Quantitative researcher (Apprenticeship)**, Ofi Invest Asset Management 2022 - 2023
- Quantitative research projects for investment solutions : Study of financial news using NLP algorithms by combining Name Entity Recognition (NER), Few-Shot Learning and Aspect Based Sentiment Analysis (ABSA) methods.

Publications

- [1] Pierre Fihey, Matthieu Labeau, and Pavlo Mozharovskiy. "A Geometry-Aware Depth Function for Robust Unsupervised Textual Anomaly Detection". In: *Findings of the Association for Computational Linguistics: EMNLP 2026 (to be published)*. 2026.
- [2] Pierre Fihey, Matthieu Labeau, and Pavlo Mozharovskiy. "Enhancing Two Steps Textual Anomaly Detection through Anisotropy Mitigation". In: *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2026.

Talks

- **ACL 2026 - Paper Presentation (Oral)** - *Enhancing Two Steps Textual Anomaly Detection through Anisotropy Mitigation*